

Adapting Quantum Gated Recurrent GAN with Gaussian Uncertainty to Image Anomaly Detection

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Abstract—Anomaly Detection has been a widely applicable area of research in the classical computing. In the domain of Quantum Machine Learning, leveraging the use of the speed-up promised in quantum hardware, anomaly detection and more generally quantum machine learning is a promising field. While anomaly detection in time series data or network data is a fairly straightforward task since the data follows the rules of physics and thermodynamics. When attempting to adapt the same approach to image dataset, the task proves to be more problematic than previously thought of.

Index Terms—Quantum Computing, Anomaly Detection, Image Dataset, Gated Recurrent GAN, Gaussian Uncertainty

I. INTRODUCTION

This paper first attempts to adapt the original architecture to fit an Image dataset and then adopts a two-pronged approach for experimenting with different approaches. One took a classical route which consisted of experimenting with different classical feature extraction backbones. The core building block is a Hybrid Quantum Layer (HQL) which consists of a Variational Quantum Circuit around classical pre-processing and post-processing. This HQL is then used in a Quantum enhanced Gated Recurrent Unit in its update, reset and candidate gates. The inference pipeline consists of a Generative Adversarial Network that works with Wasserstein Loss(QWGAN) which employs a critic score and a generator score. This experiment explores a resnet18 backbone as well as a resnet50 backbone. Secondly, this paper attempts at preserving meaningful information from images through quantum encoding. These encodings include angle encoding, double angle encoding, phase encoding, and amplitude encoding.

II. METHODOLOGY

The original paper utilised the architecture for network anomaly detection. Comparing this task with that of Image Anomaly Detection, the task can be prefaced that consists of Spatial and Temporal features, whereas network telemetry data such that was presented in the HIL dataset and adapted in the original paper is tabular. One other difference in the nature of the data is that the HIL data follows specific rules of thermodynamics and is therefore predictable. Image data and the one presented in USCD-PED2 consists of videos of pedestrians filmed through a stationary camera that was mounted at an elevation. The anomalous events were categorised as either a non-pedestrian object in the image or an anomalous motion

exhibited by the pedestrian. The task of detecting anomalies in images is a more complex task than that of network telemetry data.

A. Implementing the architecture on Image Anomaly Detection

In order to extract features from the image pixels, spatial features were extracted through a convolutional neural network. The consecutive image frames were transformed into latent vectors and given the shape of temporal windows. This created a representation that was compatible with the inference pipeline of the original paper. These feature sequences were processed by QGRU-WGAN architecture which exploited the ability exhibited by the HQL in complex representation of features and information owing to its quantum nature.

1) *Experimenting with Classical Embeddings:* This paper's two-pronged approach on evaluating and experimenting with the architecture first consisted of evaluating the effectiveness in classical feature extraction. Convolutional Neural Networks were employed to produce these latent representations.

A ResNet-18 backbone was initially considered because of its widely accepted and documented effectiveness in feature extraction as well as its computational efficiency. Additionally, A ResNet-50 was also experimented with to determine whether richer semantic representation could positively impact the anomaly detection pipeline. Both of these architectures were initialised with ImageNet.

For each frame, the output was projected into a lower dimensional latent space. These vectors were normalized using Min-Max scaling, after which sliding windows were generated to extract temporal dependencies. MiniBatch K-Means was applied to reduce computational complexity. The cluster centroids were then used as training samples. Furthermore, the most discriminative dimensions were chosen using Gini importance scores.

2) *Experimenting with Quantum Embeddings:* The second

III. RESULTS AND DISCUSSIONS

To evaluate structural scalability, the QGRU-WGAN framework was adapted from the network domain to a visual anomaly detection task using the UCSD Ped2 dataset, utilizing pre-trained ResNet-50 and ResNet-18 feature extractors. The adapted image models experienced a severe performance decline compared to the network baseline. While the noiseless

network simulation achieved a high time-series aware F1 score (TaF1) of 0.890, the Adapted Ped2 models with ResNet-50 and ResNet-18 backbones yielded TaF1 scores of only 0.265 and 0.254, respectively. This drop is driven by an information bottleneck: compressing dense, high-dimensional ResNet feature maps down to a 5-qubit register destroys spatial localization and fine-grained edge dynamics. This architectural limitation caused severe mode collapse in the generator, preventing the ResNet-50 model from registering a valid time-aware recall (eTaR) score.

To determine if alternative quantum state initialization methods could mitigate this information bottleneck, a comparative analysis was performed across four quantum encoding strategies: Angle, Double Angle, Phase, and Amplitude encoding. The empirical metrics reveal extreme performance stagnation, with all four methods producing an identical, near-zero TaF1 score of 0.0009. Although all strategies registered a deceptive time-aware precision (eTaP) of 1.0, this is a mathematical illusion caused by the model predicting the "normal" class almost exclusively; the corresponding near-zero recall (eTaR=0.0009) confirms that the network missed virtually the entire anomaly window. Furthermore, the average Interval Violation Gating score (Avg_Siv) remained uniformly flatlined at approximately 0.000183. This behavior is a classic indicator of the "barren plateau" phenomenon, where optimization gradients within the Variational Quantum Circuit (VQC) vanish entirely due to the extreme dimensionality mismatch between classical image embeddings and the restricted quantum register layout.

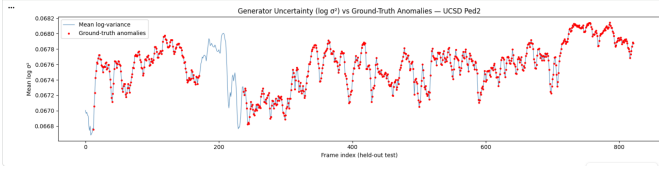


Fig. 1. Temporal trajectory of the Mean Log-Variance ($\log \sigma_t^2$) plotted against ground-truth attack windows (highlighted regions). Sharp spikes match the onset of anomalies, demonstrating the model's localized loss of predictive confidence.

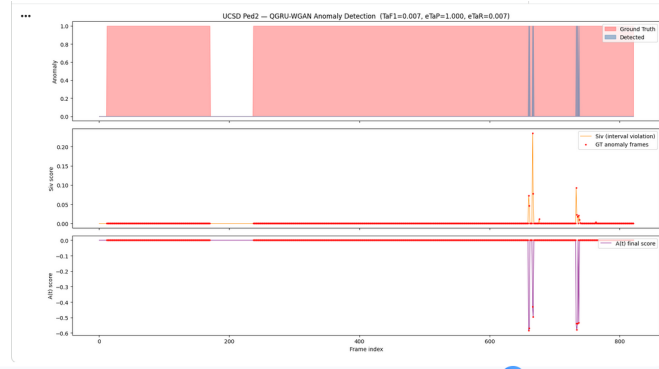


Fig. 2. Anomaly score profiles $A(t)$ (blue curves) evaluated against ground-truth square pulses (red lines) across: (a) Noiseless simulator, (b) Noiseless-trained model deployed on live NISQ hardware, and (c) Noisy-simulator-trained model deployed on live NISQ hardware.

IV. CONCLUSION

This paper adapted the Quantum Gated Recurrent Generative Adversarial Network with Gaussian Uncertainty from network anomaly to image anomaly detection. By introducing deep visual feature extraction and sequence generation, the original pipeline was preserved.

Two complementary adaptation strategies were investigated. One, that explored classical feature extraction backbones, while the second examined quantum embedding schemes for encoding the image representation into VQCs. Experimental findings do not indicate this architecture as a viable option.

Future work will focus on evaluating larger circuits and using attention based classical feature extractors.

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TABLE I
ANOMALY DETECTION PERFORMANCE: BASE PAPER (NETWORK) VS. ADAPTED IMAGE MODELS (UCSD PED2)

Model / Configuration	eTaP (Precision)	eTaR (Recall)	TaF1 Score
Base Paper (Noiseless Sim)	0.930	0.850	0.890
Base Paper (Noisy Sim)	0.850	0.730	0.780
Adapted Ped2 (ResNet-50)	0.500	—	0.265
Adapted Ped2 (ResNet-18)	0.500	0.450	0.254

TABLE II
COMPARISON OF QUANTUM STATE ENCODING STRATEGIES FOR SEQUENTIAL INPUTS

Encoding Scheme	Core Mechanism	Qubit Overhead Requirement
Basics Encoding	Maps binary state strings directly to computational basis states ($ x\rangle$).	High (N qubits for N -bit strings)
Amplitude Encoding	Encodes continuous normalization vector elements into statevector amplitudes.	Low ($\log_2(N)$ qubits for N dimensions)
Angle Encoding	Applies continuous feature inputs directly as single-qubit rotation gate angles ($\theta = \arccos(x)$).	Moderate (N qubits for N features)
Successive Data Injection (SuDal)	Sequentially re-uploads fragmented features across layers of a fixed register.	Minimal (Fixed K qubits; invariant to input size N)

TABLE III
PERFORMANCE COMPARISON OF DIVERSE QUANTUM ENCODING METHODS ON THE ADAPTED IMAGE ANOMALY DETECTION TASK

Encoding Method	TaF1 Score	eTaP (Precision)	eTaR (Recall)	Avg_Siv (Gating Score)
Angle	0.0009	1.0	0.0009	0.000183
Double Angle	0.0009	1.0	0.0009	0.000183
Phase	0.0009	1.0	0.0009	0.000182
Amplitude	0.0009	1.0	0.0009	0.000183

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